

DATA CLEANING CHANGE LOG

Project 1

Data Cleaning and Preparation
DecodeLabs Industrial Training, Batch 2026

AUDIT SUMMARY

Metric	Before	After	Status
Totatl Records	1200	1200	No Loss
Missing Values (CouponCode)	309	0	Resolved
Duplicate Rows	0	0	Clean
Duplicate Order IDs	0	0	Clean
Data Format	Mixed	YYYY-MM-DD (100%)	Resolved
Numeric Precision	Variable	2 Decimal Places	Resolved
Text Case	Mixed	Title Case (100%)	Resolved

DETAILED CHANGE RECORDS

CR001	Phase 1 – Strategic Imputation	Column(s): CouponCode	Resolved	
Issue: 309 null values found in CouponCode. These are legitimate orders where customers did not apply a coupon, not data entry errors.				
Action Taken: Filled all 309 nulls with the constant string "No Coupon". Avoided deletion to preserve statistical power.				
Impact: Preserved all 1,200 records. Missing rate reduced from 25.75% to 0%.				

CR002	Phase 2 – Integrity Audit	Column(s): OrderID / All Columns	Verified Clean	
Issue: Full duplicate audit performed. Checked for fully duplicate rows and duplicate Order IDs separately.				
Action Taken: Applied exact-match comparison across all 14 columns and a separate uniqueness check on OrderID.				
Impact: 0 duplicate rows and 0 duplicate Order IDs confirmed. All 1,200 IDs are unique.				

CR003	Phase 3 – Format Standardisation	Column(s): Date	Resolved	
Issue: Date column contained mixed datetime objects without a guaranteed consistent string format.				
Action Taken: Applied pd.to_datetime() then strftime("%Y-%m-%d") to enforce ISO 8601 format across all 1,200 rows.				
Impact: All 1,200 dates are now in YYYY-MM-DD format. Zero formatting errors.				

CR004	Phase 3 – Format Standardisation	Column(s): Product, PaymentMethod OrderStatus ReferralSource CouponCode	Resolved	
Issue: Categorical text columns risked mixed casing or stray whitespace that could cause incorrect group-by results in analysis.				
Action Taken: Applied str.strip() to remove whitespace and str.title() for uniform Title Case on all 5 text columns.				
Impact: All categorical values standardised. No case-mismatch grouping errors possible.				

CR005	Phase 3 – Format Standardisation	Column(s): UnitPrice TotalPrice	Resolved	
Issue: Monetary columns had variable decimal precision, which was inconsistent for financial reporting.				
Action Taken: Applied DataFrame.round(2). Also verified TotalPrice = Quantity x UnitPrice for all 1,200 rows.				
Impact: Consistent 2-decimal precision. Zero calculation mismatches found.				

VERIFICATION GATE, PROJECT 2 THRESHOLD

Threshold Check	Required	Result	Gate
Duplicate Order IDs	0%	0 Found	PASS
Incorrectly Formatted Dates	0%	0 Found	PASS
Missing Values Remaining	0%	0 Found	PASS

All Project 2 thresholds met. Dataset is production-ready